

Groups (2022–2026)

Category: Linear Algebra and Algebra • Official Mock Test Series

TOTAL QUESTIONS

35

TOTAL MARKS

57 M

TEST DURATION

105 Min

PAPER PATTERN

MCQ • MSQ • NAT
(2022–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	19	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	6	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	10	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2026 • Q47 • ID: JAM_2026_Q47)

NAT

+1 M

Neg: Nil (0)

Q.47	Let C_n denote a cyclic group having n elements. If there is a surjective group homomorphism from C_n to C_{30} , then the total number of such distinct surjective homomorphisms is _____.

Question 2 (JAM 2026 • Q30 • ID: JAM_2026_Q30)

MCQ

+2 M

Neg: -0.67

Q.30	Let G be the power set of $\{1,2,3,4\}$. Let the binary operation Δ on G be $A \Delta B = (A \setminus B) \cup (B \setminus A).$ Which ONE of the following statements is TRUE?
(A)	$\{1\}$ is an identity element of (G, Δ) .
(B)	(G, Δ) is an abelian group but not cyclic.
(C)	(G, Δ) is a group and has an element of order 4.
(D)	(G, Δ) is a group and has an element of order 8.

Question 3 (JAM 2026 • Q27 • ID: JAM_2026_Q27)

MCQ

+2 M

Neg: -0.67

Q.27	Which ONE of the following statements is FALSE?
(A)	There is a surjective group homomorphism from the additive group of rational numbers to the multiplicative group of all complex roots of unity.
(B)	The multiplicative group of complex numbers of modulus one is isomorphic to a quotient group of the additive group of real numbers.
(C)	Any group homomorphism from the multiplicative group of nonzero complex numbers into the group of all invertible 2×2 matrices with real entries has nontrivial kernel.
(D)	There exists a group homomorphism from the symmetric group on n symbols into the multiplicative group of all invertible $n \times n$ matrices with real entries, which has a trivial kernel.

Question 4 (JAM 2026 • Q26 • ID: JAM_2026_Q26)

MCQ

+2 M

Neg: -0.67

Q.26	Let $f: G \rightarrow \mathbb{Z}$ be a surjective group homomorphism from an abelian group G to the additive group of integers $\mathbb{Z}$. Let K denote the kernel of f. Which ONE of the following statements is FALSE?
(A)	There exists a group homomorphism $\phi: \mathbb{Z} \rightarrow G$ such that $f \circ \phi$ is the identity homomorphism.
(B)	K is isomorphic to a quotient group of G .
(C)	For all non-zero homomorphisms $\alpha: \mathbb{Z} \rightarrow G$ and $\beta: G \rightarrow K$, the composition $\beta \circ \alpha$ is non-zero.
(D)	G has a subgroup isomorphic to $\mathbb{Z}$.

Question 5 (JAM 2026 • Q9 • ID: JAM_2026_Q9)

MCQ

+1 M

Neg: -0.33

Q.9	Which ONE of the following statements is FALSE?
(A)	S_3 is isomorphic to a subgroup of S_4 .
(B)	$\mathbb{Z}_3$ is isomorphic to a subgroup of S_4 .
(C)	S_3 is isomorphic to a quotient group of S_4 .
(D)	$\mathbb{Z}_6$ is isomorphic to a quotient group of S_4 .

Question 6 (JAM 2025 • Q60 • ID: JAM_2025_Q60)

NAT

+2 M

Neg: Nil (0)

Q.60 The number of surjective (onto) group homomorphisms from S_4 to $\mathbb{Z}_6$ is equal to _____.

Question 7 (JAM 2025 • Q59 • ID: JAM_2025_Q59)

NAT

+2 M

Neg: Nil (0)

Q.59 Let G be an abelian group of order 35. Let m denote the number of elements of order 5 in G , and let n denote the number of elements of order 7 in G .
Then, the value of $m + n$ is equal to _____.

Question 8 (JAM 2025 • Q52 • ID: JAM_2025_Q52)

NAT

+2 M

Neg: Nil (0)

Q.52 Let $\sigma \in S_4$ be the permutation defined by $\sigma(1) = 2$, $\sigma(2) = 3$, $\sigma(3) = 1$ and $\sigma(4) = 4$. The number of elements in the set $\{\tau \in S_4 : \tau\sigma\tau^{-1} = \sigma\}$ is equal to _____.

Question 9 (JAM 2025 • Q39 • ID: JAM_2025_Q39)

MSQ

+2 M

Neg: Nil (0)

Q.39 For $n \in \mathbb{N}$, consider the set $U(n) = \{\bar{x} \in \mathbb{Z}_n : \gcd(x, n) = 1\}$ as a group under multiplication modulo n .

Then, which of the following is/are TRUE?

- (A) $U(8)$ is a cyclic group
- (B) $U(5)$ is a cyclic group
- (C) $U(12)$ is a cyclic group
- (D) $U(9)$ is a cyclic group

Question 10 (JAM 2025 • Q23 • ID: JAM_2025_Q23)

MCQ

+2 M

Neg: -0.67

Q.23 Let $\mathbb{R}/\mathbb{Z}$ denote the quotient group, where $\mathbb{Z}$ is considered as a subgroup of the additive group of real numbers $\mathbb{R}$.

Let m denote the number of injective (one-one) group homomorphisms from $\mathbb{Z}_3$ to $\mathbb{R}/\mathbb{Z}$ and n denote the number of group homomorphisms from $\mathbb{R}/\mathbb{Z}$ to $\mathbb{Z}_3$.

Then, which one of the following is TRUE?

- (A) $m = 2$ and $n = 1$
- (B) $m = 3$ and $n = 3$
- (C) $m = 2$ and $n = 3$
- (D) $m = 1$ and $n = 1$

Question 11 (JAM 2025 • Q9 • ID: JAM_2025_Q9)

MCQ

+1 M

Neg: -0.33

Q.9 Let G be a finite abelian group of order 10. Let x_0 be an element of order 2 in G .

If $X = \{x \in G : x^3 = x_0\}$, then which one of the following is TRUE?

- (A) X has exactly one element
- (B) X has exactly two elements
- (C) X has exactly three elements
- (D) X is an empty set

Question 12 (JAM 2024 • Q40 • ID: JAM_2024_Q40)

MSQ

+2 M

Neg: Nil (0)

Q.40 Consider

$$G = \{m + n\sqrt{2} : m, n \in \mathbb{Z}\}$$

as a subgroup of the additive group $\mathbb{R}$.

Which of the following statements is/are TRUE?

- (A) G is a cyclic subgroup of $\mathbb{R}$ under addition
- (B) $G \cap I$ is non-empty for every non-empty open interval $I \subseteq \mathbb{R}$
- (C) G is a closed subset of $\mathbb{R}$
- (D) G is isomorphic to the group $\mathbb{Z} \times \mathbb{Z}$, where the group operation in $\mathbb{Z} \times \mathbb{Z}$ is defined by $(m_1, n_1) + (m_2, n_2) = (m_1 + m_2, n_1 + n_2)$

Question 13 (JAM 2024 • Q36 • ID: JAM_2024_Q36)

MSQ

+2 M

Neg: Nil (0)

Q.36 The center $Z(G)$ of a group G is defined as

$$Z(G) = \{x \in G : xg = gx \text{ for all } g \in G\}.$$

Let $|G|$ denote the order of G . Then, which of the following statements is/are TRUE for any group G ?

- (A) If G is non-abelian and $Z(G)$ contains more than one element, then the center of the quotient group $G/Z(G)$ contains only one element
- (B) If $|G| \geq 2$, then there exists a non-trivial homomorphism from $\mathbb{Z}$ to G
- (C) If $|G| \geq 2$ and G is non-abelian, then there exists a non-identity isomorphism from G to itself
- (D) If $|G| = p^3$, where p is a prime number, then G is necessarily abelian

Question 14 (JAM 2024 • Q33 • ID: JAM_2024_Q33)

MSQ

+2 M

Neg: Nil (0)

Q.33 Which of the following statements is/are TRUE?

- (A) The additive group of real numbers is isomorphic to the multiplicative group of positive real numbers
- (B) The multiplicative group of nonzero real numbers is isomorphic to the multiplicative group of nonzero complex numbers
- (C) The additive group of real numbers is isomorphic to the multiplicative group of nonzero complex numbers
- (D) The additive group of real numbers is isomorphic to the additive group of rational numbers

Question 15 (JAM 2024 • Q12 • ID: JAM_2024_Q12)

MCQ

+2 M

Neg: -0.67

Q.12 Consider the group $G = \{A \in M_2(\mathbb{R}) : AA^T = I_2\}$ with respect to matrix multiplication. Let

$$Z(G) = \{A \in G : AB = BA, \text{ for all } B \in G\}.$$

Then, the cardinality of $Z(G)$ is

- (A) 1
- (B) 2
- (C) 4
- (D) Infinite

Question 16 (JAM 2024 • Q11 • ID: JAM_2024_Q11)

MCQ

+2 M

Neg: -0.67

Q.11 Which one of the following groups has elements of order 1, 2, 3, 4, 5 but does not have an element of order greater than or equal to 6 ?

- (A) The alternating group A_6
- (B) The alternating group A_5
- (C) S_6
- (D) S_5

Question 17 (JAM 2024 • Q8 • ID: JAM_2024_Q8)

MCQ

+1 M

Neg: -0.33

Q.8 Let G be a finite group containing a non-identity element which is conjugate to its inverse. Then, which one of the following is TRUE?

- (A) The order of G is necessarily even
- (B) The order of G is not necessarily even
- (C) G is necessarily cyclic
- (D) G is necessarily abelian but need not be cyclic

Question 18 (JAM 2024 • Q7 • ID: JAM_2024_Q7)

MCQ

+1 M

Neg: -0.33

Q.7 Which one of the following is TRUE for the symmetric group S_{13} ?

- (A) S_{13} has an element of order 42
- (B) S_{13} has no element of order 35
- (C) S_{13} has an element of order 27
- (D) S_{13} has no element of order 60

Question 19 (JAM 2024 • Q6 • ID: JAM_2024_Q6)

MCQ

+1 M

Neg: -0.33

Q.6 For a positive integer n , let $U(n) = \{\bar{r} \in \mathbb{Z}_n : \gcd(r, n) = 1\}$ be the group under multiplication modulo n . Then, which one of the following statements is TRUE?

- (A) $U(5)$ is isomorphic to $U(8)$
- (B) $U(10)$ is isomorphic to $U(12)$
- (C) $U(8)$ is isomorphic to $U(10)$
- (D) $U(8)$ is isomorphic to $U(12)$

Question 20 (JAM 2024 • Q5 • ID: JAM_2024_Q5)

MCQ

+1 M

Neg: -0.33

Q.5 Let G be a group of order 39 such that it has exactly one subgroup of order 3 and exactly one subgroup of order 13. Then, which one of the following statements is TRUE?

- (A) G is necessarily cyclic
- (B) G is abelian but need not be cyclic
- (C) G need not be abelian
- (D) G has 13 elements of order 13

Question 21 (JAM 2023 • Q54 • ID: JAM_2023_Q54)

NAT

+2 M

Neg: Nil (0)

Q. 54 The number of permutations in S_4 that have exactly two cycles in their cycle decompositions is equal to _____.

Question 22 (JAM 2023 • Q46 • ID: JAM_2023_Q46)

NAT

+1 M

Neg: Nil (0)

Q. 46 For $g \in \mathbb{Z}$, let $\bar{g} \in \mathbb{Z}_8$ denote the residue class of g modulo 8. Consider the group $\mathbb{Z}_8^\times = \{\bar{x} \in \mathbb{Z}_8 : 1 \leq x \leq 7, \gcd(x, 8) = 1\}$ with respect to multiplication modulo 8. The number of group isomorphisms from $\mathbb{Z}_8^\times$ onto itself is equal to _____.

Question 23 (JAM 2023 • Q45 • ID: JAM_2023_Q45)

NAT

+1 M

Neg: Nil (0)

Q. 45 For $\sigma \in S_8$, let $o(\sigma)$ denote the order of σ . Then $\max\{o(\sigma) : \sigma \in S_8\}$ is equal to _____.

Question 24 (JAM 2023 • Q32 • ID: JAM_2023_Q32)

MSQ

+2 M

Neg: Nil (0)

Q. 32 Let $A \subseteq \mathbb{Z}$ with $0 \in A$. For $r, s \in \mathbb{Z}$, define

$$rA = \{ra : a \in A\}, \quad rA + sA = \{ra + sb : a, b \in A\}.$$

Which of the following conditions imply that A is a subgroup of the additive group $\mathbb{Z}$?

- (A) $-2A \subseteq A, \quad A + A = A$
- (B) $A = -A, \quad A + 2A = A$
- (C) $A = -A, \quad A + A = A$
- (D) $2A \subseteq A, \quad A + A = A$

Question 25 (JAM 2023 • Q29 • ID: JAM_2023_Q29)

MCQ

+2 M

Neg: -0.67

Q. 29 From the additive group $\mathbb{Q}$ to which one of the following groups does there exist a non-trivial group homomorphism?

- (A) $\mathbb{R}^\times$, the multiplicative group of non-zero real numbers
- (B) $\mathbb{Z}$, the additive group of integers
- (C) $\mathbb{Z}_2$, the additive group of integers modulo 2
- (D) $\mathbb{Q}^\times$, the multiplicative group of non-zero rational numbers

Question 26 (JAM 2023 • Q27 • ID: JAM_2023_Q27)

MCQ

+2 M

Neg: -0.67

Q. 27 How many group homomorphisms are there from $\mathbb{Z}_2$ to S_5 ?

- (A) 40
- (B) 41
- (C) 26
- (D) 25

Question 27 (JAM 2023 • Q23 • ID: JAM_2023_Q23)

MCQ

+2 M

Neg: -0.67

Q. 23 Consider the following statements:

- I. Every infinite group has infinitely many subgroups.
- II. There are only finitely many non-isomorphic groups of a given finite order.

Then

- (A) both I and II are TRUE
- (B) I is TRUE but II is FALSE
- (C) I is FALSE but II is TRUE
- (D) both I and II are FALSE

Question 28 (JAM 2023 • Q1 • ID: JAM_2023_Q1)

MCQ

+1 M

Neg: -0.33

Q. 1 Let G be a finite group. Then G is necessarily a cyclic group if the order of G is

- (A) 4
- (B) 7
- (C) 6
- (D) 10

Question 29 (JAM 2022 • Q58 • ID: JAM_2022_Q58)

NAT

+2 M

Neg: Nil (0)

Q.58

Let σ be the permutation in the symmetric group S_5 given by

$$\sigma(1) = 2, \quad \sigma(2) = 3, \quad \sigma(3) = 1, \quad \sigma(4) = 5, \quad \sigma(5) = 4.$$

Define

$$N(\sigma) = \{\tau \in S_5 : \sigma \circ \tau = \tau \circ \sigma\}.$$

Then the number of elements in $N(\sigma)$ is equal to _____.

Question 30 (JAM 2022 • Q46 • ID: JAM_2022_Q46)

NAT

+1 M

Neg: Nil (0)

Q.46

The number of elements of order 12 in the symmetric group S_7 is equal to

_____.

Question 31 (JAM 2022 • Q45 • ID: JAM_2022_Q45)

NAT

+1 M

Neg: Nil (0)

Q.45

The number of distinct subgroups of $\mathbb{Z}_{999}$ is _____.

Question 32 (JAM 2022 • Q37 • ID: JAM_2022_Q37)

MSQ

+2 M

Neg: Nil (0)

Q.37	Let G be a finite group of order at least two and let e denote the identity element of G. Let $\sigma: G \rightarrow G$ be a bijective group homomorphism that satisfies the following two conditions: (i) If $\sigma(g) = g$ for some $g \in G$, then $g = e$, (ii) $(\sigma \circ \sigma)(g) = g$ for all $g \in G$. Then which of the following is/are correct?
(A)	For each $g \in G$, there exists $h \in G$ such that $h^{-1}\sigma(h) = g$.
(B)	There exists $x \in G$ such that $x\sigma(x) \neq e$.
(C)	The map σ satisfies $\sigma(x) = x^{-1}$ for every $x \in G$.
(D)	The order of the group G is an odd number.

Question 33 (JAM 2022 • Q24 • ID: JAM_2022_Q24)

MCQ

+2 M

Neg: -0.67

Q.24	For $g \in \mathbb{Z}$, let $\bar{g} \in \mathbb{Z}_{37}$ denote the residue class of g modulo 37. Consider the group $U_{37} = \{\bar{g} \in \mathbb{Z}_{37} : 1 \leq g \leq 37 \text{ with } \gcd(g, 37) = 1\}$ with respect to multiplication modulo 37. Then which one of the following is FALSE?
(A)	The set $\{\bar{g} \in U_{37} : \bar{g} = (\bar{g})^{-1}\}$ contains exactly 2 elements.
(B)	The order of the element $\bar{10}$ in U_{37} is 36.
(C)	There is exactly one group homomorphism from U_{37} to $(\mathbb{Z}, +)$.
(D)	There is exactly one group homomorphism from U_{37} to $(\mathbb{Q}, +)$.

Question 34 (JAM 2022 • Q19 • ID: JAM_2022_Q19)

MCQ

+2 M

Neg: -0.67

Q.19	Consider the group $(\mathbb{Q}, +)$ and its subgroup $(\mathbb{Z}, +)$. For the quotient group $\mathbb{Q}/\mathbb{Z}$, which one of the following is FALSE?
(A)	$\mathbb{Q}/\mathbb{Z}$ contains a subgroup isomorphic to $(\mathbb{Z}, +)$.
(B)	There is exactly one group homomorphism from $\mathbb{Q}/\mathbb{Z}$ to $(\mathbb{Q}, +)$.
(C)	For all $n \in \mathbb{N}$, there exists $g \in \mathbb{Q}/\mathbb{Z}$ such that the order of g is n .
(D)	$\mathbb{Q}/\mathbb{Z}$ is not a cyclic group.

Question 35 (JAM 2022 • Q4 • ID: JAM_2022_Q4)

MCQ

+1 M

Neg: -0.33

Q.4	Let G be a group of order 2022. Let H and K be subgroups of G of order 337 and 674, respectively. If $H \cup K$ is also a subgroup of G , then which one of the following is FALSE?
(A)	H is a normal subgroup of $H \cup K$.
(B)	The order of $H \cup K$ is 1011.
(C)	The order of $H \cup K$ is 674.
(D)	K is a normal subgroup of $H \cup K$.

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Groups (2022–2026) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key	Q. No.	Type	Marks	Official Key	Q. No.	Type	Marks	Official Key
1	NAT	+1	8 to 8	13	MSQ	+2	B;C	25	MCQ	+2	A
2	MCQ	+2	B	14	MSQ	+2	A	26	MCQ	+2	C
3	MCQ	+2	C	15	MCQ	+2	B	27	MCQ	+2	A
4	MCQ	+2	C	16	MCQ	+2	A	28	MCQ	+1	B
5	MCQ	+1	D	17	MCQ	+1	A	29	NAT	+2	6 to 6
6	NAT	+2	0 to 0	18	MCQ	+1	A	30	NAT	+1	420 to 420
7	NAT	+2	10 to 10	19	MCQ	+1	D	31	NAT	+1	8 to 8
8	NAT	+2	3 to 3	20	MCQ	+1	A	32	MSQ	+2	A,C,D
9	MSQ	+2	B;D	21	NAT	+2	11 to 11	33	MCQ	+2	B
10	MCQ	+2	A	22	NAT	+1	6 to 6	34	MCQ	+2	A
11	MCQ	+1	A	23	NAT	+1	15 to 15	35	MCQ	+1	B
12	MSQ	+2	B;D	24	MSQ	+2	A, B, C				

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.